C6.3 What factors affect the yield of chemical reactions?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	***Linked learning opportunities***
Industrial processes are managed to get the best yield as quickly and economically as possible. Chemists select the conditions that give the best economic outcome in terms of safety, maintaining the conditions and equipment, and energy use. The reactions in some processes are reversible. This can be problematic in industry because the reactants never completely react to make the products. This wastes reactants and means that the products have to be separated out from the reactants, which requires extra stages and costs. **Data about yield and rate of chemical processes are used to choose the best conditions to make a product. On industrial scales, very high temperatures and pressures are expensive to maintain due to the cost of energy and because equipment may fail under extreme conditions. Catalysts can be used to increase the rate of reaction without affecting yield.** Chemical engineers choose the conditions that will make the process as safe and efficient as possible, reduce the energy costs and reduce the waste produced at all stages of the process.	1. recall that some reactions may be reversed by altering the reaction conditions including: a) reversible reactions are shown by the symbol $\rightleftharpoons$ b) reversible reactions (in closed systems) do not reach 100% yield 2. recall that dynamic equilibrium occurs when the rates of forward and reverse reactions are equal **3. predict the effect of changing reaction conditions (concentration, temperature and pressure) on equilibrium position and suggest appropriate conditions to produce a particular product, including:** **a) catalysts increase rate but do not affect yield** **b) the disadvantages of using very high temperatures or pressures**	**Specification links:** • Calculations of yields (C5.1) **Practical Work:** • Investigating reversible reactions **Ideas about Science:** • Make predictions from data and graphs about yield of chemical products. (IaS2) • Consider the risks and costs of different operating conditions in an ammonia plant. (IaS4) **Practical work** • Analyse and evaluate data about yield and rate of ammonia production.